Reg. No.														

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, JULY 2025
Third Semester

21MAB201T – TRANSFORMS AND BOUNDARY VALUE PROBLEMS
(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer **ALL** Questions

Marks BL CO

1. The partial differential equation formed by eliminating arbitrary constant is $z=(x+a)(y+b)$ — 1 1 1
   (A) $z=p+q$ (B) $z=p-q$
   (C) $z=\frac{p}{q}$ (D) $z=pq$

2. The general integral of $z=xp+yq$ is — 1 1 1
   (A) $\Phi(x+y,y+z)=0$ (B) $\Phi\left(\frac{x}{y}+\frac{y}{z}\right)=0$
   (C) $\Phi\left(x-y,\frac{x}{z}\right)=0$ (D) $\Phi\left(\frac{x}{y},y+z\right)=0$

3. The particular integral of $(D^3-2D^2D')z=e^{x+2y}$ is — 1 2 1
   (A) $\frac{e^{x+2y}}{3}$ (B) $\frac{e^{x}}{3}$
   (C) e^{x+2y} (D) $\frac{-e^{x+2y}}{3}$

4. Solve $(D^3-7DD'^2-6D'^3)z=0$ — 1 2 1
   (A) $z=f_1(y-x)+f_2(y-2x)+f_3(y+3x)$
   (B) $z=f_1(y-x)+f_2(y+2x)+f_3(y-3x)$
   (C) $z=f_1(y+x)+f_2(y-2x)+f_3(y+3x)$
   (D) $z=f_1(y+x)+f_2(y+2x)+f_3(y+3x)$

5. tan x is a periodic function with period — 1 1 1
   (A) 2π (B) π
   (C) 3π (D) π/2

6. The constant a_0 of the Fourier series for the function f(x) = k, $0\le x\le 2\pi$ is — 1 2 2
   (A) k (B) 2k
   (C) 0 (D) k/2

7. The RMS value of f(x) in $a \le x \le b$ is 1 2 2

(A) $\sqrt{\dfrac{\int_a^b [f(x)]^2\,dx}{b-a}}$ (B) 0

(C) $\sqrt{\dfrac{\int_a^b [f(x)]^2\,dx}{b+a}}$ (D) $\sqrt{\dfrac{\sqrt{\int_a^b f(x)dx}}{b-a}}$

8. For half range cosine series of $f(x) = \cos x$ in $(0,\pi)$ the value of a_0 is 1 2 2

(A) 4 (B) $\dfrac{2}{\pi}$

(C) $\dfrac{4}{\pi}$ (D) 0

9. How many initial and boundary conditions are required to solve $\dfrac{\partial^2 y}{\partial t^2} = a^2 \dfrac{\partial^2 y}{\partial x^2}$ 1 2 3

(A) Two (B) Three

(C) Five (D) Four

10. In heat equation $\dfrac{\partial u}{\partial t} = \alpha^2 \dfrac{\partial^2 u}{\partial x^2}, \alpha^2$ stands for 1 1 3

(A) $\dfrac{k}{p}$ (B) $\dfrac{T}{m}$

(C) $\dfrac{k}{pc}$ (D) $\dfrac{k}{c}$

11. One dimensional wave equation is used to find 1 1 3

(A) Temperature (B) Displacement

(C) Time (D) Mass

12. The steady state temperature of a rod of length. Whose ends are kept at 30° and 40° is 1 2 3

(A) $u = \dfrac{10x}{l} + 30$ (B) $u = \dfrac{20x}{l} + 30$

(C) $u = \dfrac{10x}{l} + 20$ (D) $u = \dfrac{10x}{l}$

13. The Fourier cosine transform of e^{-ax} is 1 1 4

(A) $\sqrt{\dfrac{2}{\pi}} \dfrac{a}{a^2+x^2}$ (B) $\sqrt{\dfrac{1}{\pi}} \dfrac{s}{s^2+a^2}$

(C) $\sqrt{\dfrac{1}{\pi}} \dfrac{a}{s^2+a^2}$ (D) $\sqrt{\dfrac{2}{\pi}} \dfrac{a}{s^2+a^2}$

14. $F\left[f(x)\cos ax\right] =$ 1 1 4

(A) $\dfrac{1}{2}\left[F(a)+F(s-a)\right]$ (B) $\dfrac{1}{2}\left[F(as)+F(s+a)\right]$

(C) $\dfrac{1}{2}\left[F(s+a)+F(s-a)\right]$ (D) $\dfrac{1}{2}\left[F(s+a)-F(s-a)\right]$

15. $F\left[e^{iax} f(x)\right] =$ 1 1 4

(A) F(s+a)
(B) F(s-a)
(C) F(sa)
(D) F(s/a)

16. $F\left[f(x) * g(x)\right] =$ 1 4

(A) $F(s)+G(s)$
(B) $F(s)-G(s)$
(C) $F(s)G(s)$
(D) $F(s)/G(s)$

17. $z\left[\sin\frac{n\pi}{2}\right] =$ 1 1 5

(A) $\frac{z^2}{z^2-1}$
(B) $\frac{z}{z^2+4}$
(C) $\frac{z}{z^2+1}$
(D) $\frac{z^2}{z^2+1}$

18. Find $z\left[\frac{1}{7^n}\right]$ 1 2 5

(A) $\frac{7z}{z-1}$
(B) $\frac{7z}{7z-1}$
(C) $\frac{z}{7z-1}$
(D) $\frac{z}{z-1}$

19. $z^{-1}(e^{1/z}) =$ 1 1 5

(A) $\frac{1}{n+1}$
(B) $\frac{1}{(n+1)!}$
(C) $\frac{1}{n!}$
(D) $\frac{1}{n}$

20. Find $z^{-1}\left[\frac{z}{(z-1)^2}\right]$ 1 2 5

(A) n+1
(B) n
(C) n-1
(D) $\frac{1}{n}$

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

	Marks	BL	CO
21. a. Find the general solution $x(z^2-y^2)p + y(x^2-z^2)q = z(y^2-x^2)$.	8	3	1
(OR)			
b. Solve $(D^2-2DD')z = x^3y + e^{2x}$.	8	3	1
22. a. Expand $f(x)=(x-1)^2$ *in* $0<x<1$ in a Fourier sine series.	8	3	2

(OR)

b. Express $f(x) = x(\pi - x),\ 0 < x < \pi$, as a Fourier sine series. 8 3 2

23. a. A tightly stretched string has its ends fixed at $x = 0$ and $x = l$. Initially the string is in the form $y = kx\ (l-x)$, where k is a constant, and then released form rest. Find the displacement at any point x and any time $t > 0$. 8 4 3

(OR)

b. A rod of length l has its end A and B kept at 0°C and 100°C until steady state conditions prevail. If the temperature at B is reduced suddenly to 0°C and kept so while that of A is maintained, find the temperature $u(x,t)$ at a distance x from A and at time t. 8 4 3

24. a. Find $F_s(e^{-ax})$ and $F_c(e^{-ax})$ and hence derive the inversion formula. 8 3 4

(OR)

b. Prove that $e^{-\frac{x^2}{2}}$ is self-reciprocal under Fourier transforms. 8 3 4

25. a. Find the inverse z-transform of $\dfrac{1+2z^{-1}}{1-z^{-1}}$ by long division method. 8 3 5

(OR)

b. Find the inverse z-transform of $\dfrac{4z^2-12z}{z^3-3z+2}$ by partial fraction method. 8 3 5

PART – C (1 × 15 = 15 Marks) Marks BL CO

Answer **ANY ONE** Questions

26. Solve $y_{n+2} - 7y_{n+1} + 12y_n = 2^n$, given $y_0 = 0, y_1 = 0$ using Z transform method. 15 4 5

27. Compute first three harmonics of the half-range cosine series of y = f(x) from 15 4 2

x	0	1	2	3	4	5
f(x)	4	8	15	7	6	2

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